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DIY: Build Your Own HTPC

Kumar Jhuremalani

Ever felt cluttered with the tons of DVDs and VCDs lying around your home or on your TV? What about the interface of your home theatre system, is it too slow or not that friendly? That's where an HTPC comes into the picture.

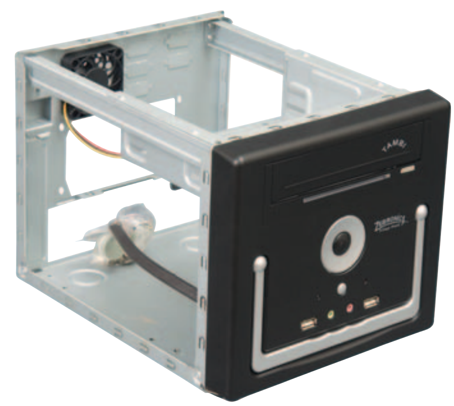
HTPC stands for home theatre personal computer or home theatre PC. Simply put, all this means, is building a PC for entertainment purposes that combines the benefits of a computer and a television integrated into one system. It's a good option, as it saves space with the capabilities of a computer. Another good thing about having your computer connected to your TV is being able to record shows on the TV directly onto your computer, especially when you are unable to watch the program while it's being broadcast. With HTPC software, all this is possible for free. Having an HTPC allows you to make slideshows, presentations, surf the web, and any other general computer task, all from the comfort of your couch — if you plan to set it up in the living room, that is. Combine this with a wireless keyboard and mouse and you're all set to enjoy an HTPC to the full.

Basic computers are normally set up for tasks such as checking mail, editing office documents, etc. An HTPC, however, does require specific hardware for home entertainment. While planning an HTPC, its cabinet needs to be specifically selected, as it needs to match the environment of your TV and living room and not look out of place. Also, remember you need something small and compact. Choosing the correct motherboard is also important as space saving is quite essential and should also have sufficient expansion slots to connect the remaining hardware. Next, you need a TV tuner card in order to watch TV on your PC. A processor with adequate speed as well as enough system RAM is a must. The prices of hard drives have also dropped, and unless you have all your movies on DVDs, you're going to need all the storage space you can get. A graphics card and a sound card are optional.

Noise is an important factor when building an HTPC. If you plan to keep the HTPC in your living room, you wouldn't want it to be noisy, lest it interfere with your work, or play. You can easily counter noise by selecting the right cabinet, choosing the right components and managing fan placement. You should have one intake and one exhaust fan, preferably silent ones. Usually, processors that generate more heat have their fans running at full clip to keep them cool, thus, processor selection is also critical.

The parts we used for building our HTPC are as follows:

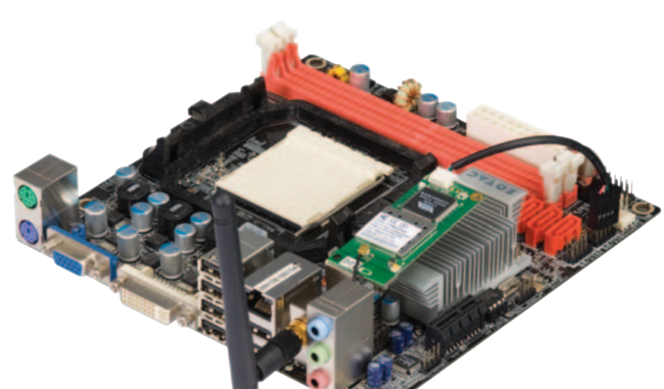
• **The case:** the HTPC case is important as this determines the ultimate look of the HTPC itself, and it will also have to fit the rest of the hardware. The case we selected is the Zebronics Tambi designed as a cube. We chose this particular case because it is compact, and has a 3.5-inch optical drive bay. The only limitation that you might face when opting for such a small cabinet is having only a single hard drive bay. However, with 1.5 TB hard drives now more affordable, this shouldn't stop you from choosing such a cabinet. Further, it should also have front USB and audio connectors for attaching headphones, or to plug a flash drive for quick access. Some cabinets available in the market are very sleek and resemble DVD players. Nevertheless, be warned - such cabinets will not have much room for expansion cards or even a 3.5-inch optical drive bay. For these cabinets, you will also need to purchase a slim drive. The reason we did not opt for this is the absence of slim Blu-ray drives.



The Zebronics Tambi mini-ITX cabinet

• **The motherboard:** we've used a mini-ITX motherboard from Zotac. You can clearly make out the size differences between an ATX, mini-ATX and mini-ITX motherboards.

The Zotac GeForce 8200 ITX-WiFi has been designed for the AMD platform and is really compact and well laid out. It has six USB ports, a VGA port and a DVI port. Zotac also supplies a DVI-to-HDMI converter with the motherboard. Since space is limited due to its size, this board has four SATA ports and two DIMMs for



Zotac GeForce 8200 ITX board with WiFi